# CLUES TO THE CLIMATE

## ICE CORES

**When an ice sheet or a glacier forms** over many years, gases from the air are trapped inside as tiny bubbles. These can provide clues about Earth's climate at different points in the past. Experts drill deep into the ice to extract long icy cylinders and examine them to find out which gases were in the atmosphere at different times in the history of our planet. This can tell us how the levels of greenhouse gases—gases that trap heat and make Earth's surface warmer—have varied over time, changing the climate.


**1 Drilling into the ice**
In the freezing Antarctic, scientists use a special drill to extract a cylindrical block of ice from the ice sheets. The oldest ice can be found in the center of an ice sheet, so the drilling must take place far from the coast and far from any research station.

### ATMOSPHERIC GASES

Earth's atmosphere is mostly made up of nitrogen and oxygen, along with small amounts of other trace gases, including helium, ozone, and carbon dioxide (a greenhouse gas). But the proportion of these gases has changed throughout history. Scientists can count layers in the ice like rings in a tree to determine how old each layer of ice is. Through this, they have discovered that since the Industrial Revolution in the 1760s, the levels of carbon dioxide in the atmosphere have been rising steadily.